# Exploring the Entanglement of Participatory Design, Spatiality and Place

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## Abstract

Over the last two decades, Participatory Design (PD) has expanded its scope towards the democratisation of design in diverse settings, embracing the entanglements of humans and non-humans in socio-material arrangements. Despite the foregrounding of relations in these entanglements, little attention has been given to what role spatiality and places may have in PD. This workshop explores the entanglement of participatory design, spatiality, technologies and places based on design cases from participants. Using a design charette methodology, participants will collaboratively (1) map how participatory entanglements shape and are shaped by design research and practice; (2) identify and unpack disciplinary tensions and opportunities; and (3) co-create a vision and action plan for bridging silos and fostering collaboration across disciplines. The aim is to build an interdisciplinary understanding of the intimate entanglement of humans and non-humans, spatiality and places in the complex social-material arrangements we engage with in PD.

## CCS Concepts

• **Human-centered computing** → Interaction design; Interaction design process and methods; Participatory design.

## Keywords

Participatory Design, Entanglement theories, Place, Place theory, Assemblage theory

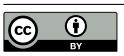
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## 1 Introduction

Over the last two decades Participatory Design (PD) has expanded its scope towards the democratisation of design in diverse settings, and research is entering a fourth wave characterized by an "increasingly intimate entanglement of humans and digital technology" ([13], p2). However, little attention has been paid to the spatial dimensions of these entanglements, a concern long shared in architecture and urban planning. The concepts of context, space, and place often overlap, and the terms space and place are frequently used interchangeably [15], a result of shifts in spatial theory during the late 20th century [20] that influenced a range of disciplines from architecture, urban planning and social psychology to ICT.

This workshop explores renewed perspectives on *space* and *place* and their intersections with technology in PD. We identify three shifts that challenge the role of PD in complex socio-material settings. *First*, design has moved from bounded contexts to open, multi-stakeholder settings, expanding PD's focus from "democracy at work" to broader citizenship and public engagement [2]. Influenced by STS and ANT, design now addresses socio-material assemblages of human and non-human actors [3, 12, 18]. *Second*, following Star and Ruhleder [22, 23] and Bowker's [5] idea of *infrastructural inversion*, attention has shifted from people and artefacts to the infrastructures that connect them, emphasizing relational dynamics and non-human actants. *Third*, agency is now viewed as distributed across humans and non-humans, informing more-than-human design [1, 14, 19]. From a cross-disciplinary perspective, Clarke et al. [6] discuss the entanglements of human and non-human worlds to counter narratives of human privilege in the urban environment. Similarly, Whatmore [24, 25] argues that the creativity of cultural geography is generated through enduring preoccupations with the processes and excesses of 'livingness' in a more-than-human world. Entanglement theories [13, 21] further develop this relational ontology, drawing on ANT, post-phenomenology, mediation theory, and new materialism. These ideas have shaped PD concepts such as infrastructuring [17], design things [12], institutioning [16], agonistic PD [4], and design as democratic inquiry [9].

Our entanglement with human and non-human elements unfolds continuously across time and space, opening up for an ontological basis for PD that links technology, society, and the spatial. Recent "place thinking" provides such grounding. For example, Kim Dovey's concept of *place-as-assemblage* [10, 11], drawing on Deleuze and Guattari's [8] and DeLanda's [7] assemblage theory, aligns with the expanded scope of PD described above. Dovey describes assemblage as a dynamic "layout" or "alignment" involving both process and spatial form. An assemblage's properties emerge from interactions among contingent parts—for example, buildings, trees, people, and signs forming a street. As these elements change, places remain fluid and continually in the process of becoming [10, 11].



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## 2 Purpose

With the expanded perspectives and scope for design and design research suggested above, there is room for development in our approaches to place and spatiality embracing non-human actors, distribution of agency, and the dynamics of relationality in PD. In this workshop, we propose to take the idea of place-as assemblage as a starting point to discuss how we could reinvigorate a concern for spaces and places to better understand the intimate entanglement of humans and non-humans in complex social-material arrangements. We would like to discuss the impacts of the expanded scope in PD described above on how we understand and engage technologies, place and spatiality in design, and how the entanglements of relations in places of design, shape and are shaped by the socio-material arrangements that we design within. The workshop's goal is to be open to and inclusive of a vast diversity of perspectives and insights originating in disciplinary differences as well as in different fields of engagement. Therefore, contributions from disciplines to be included in the workshop may span the varied fields of PD, interaction design, service design, architecture and urban design.

We aim to bring together participants with different disciplinary backgrounds to debate and pinpoint the cross disciplinary entanglements. As an example of the type of cases we have in mind, one author has applied place theory from cultural geography in studying a youth café initiative in a socially challenged suburb of a city in the Global South to foster youth social development, counter gang influence, and reduce unemployment by improving job readiness. Another author has investigated bottom-up planning and design approaches in informal settlements where communities supported by diverse institutional actors drove and enacted the construction of their area and, ultimately, their place. A third author studied the place-ness of design studio activities, suggesting that the studio as a place co-evolve as design processes unfold. Participants are asked to prepare a A3 poster for the workshop, presenting a design case, the participant has been engaged with, and pointing to what perspectives and insights have been pursued when addressing issues of place and spatiality.

## 3 Activities

### 3.1 Overview

This hands-on workshop will explore participatory design (PD) at the intersections of place, technology, and interdisciplinary collaboration. Using a design charrette methodology, participants will collaboratively: (1) Map how participatory entanglements shape and are shaped by design research and practice (with participant case studies); (2) Identify and unpack cross-disciplinary tensions and opportunities through an exploratory exercise focusing on cross-disciplinary entanglement; and (3) Co-create a vision and action plan for bridging silos and fostering collaboration across disciplines. Through hands-on activities, participants will engage in relational mapping and co-design exercises to generate insights and concrete strategies for interdisciplinary engagement.

### 3.2 Structure & activities (3 hours total)

Envisioning a maximum of 20 participant in groups of 5 people, we will encourage disciplinary diversity among groups.

**Introduction (10 min)**

- Facilitators detail the objectives, structure and activities of the session.
- Setting the Stage: PD & the Entanglement of Place, Technology, and Design - framing of the workshop

**Case study discussion (30min)**

- Participants introductions in their groups (5min)
- Walkabout to see case study posters [or small group discussion on posters depending on the notice participants have to prepare these before the conference] (25min)

**Mapping of entanglement by identifying opportunities and challenges (90min) – facilitators guide the steps**

- Groups discuss and collaborate to create a concept map of the entanglement found through the case study walkabout (members are encouraged to discuss their case studies)
- Identify common elements, the disciplines, key actors (human & non-human), technologies, places, and socio material forces
- Draw connections between these elements, considering power dynamics, agency, and interdependencies
- Annotate tensions, blind spots, or disciplinary assumptions
- Explore challenges and opportunities of these entanglements. Participants write these on Post-it notes
- Each group presents their analysis

**Co-creating the vision and plan (35min)**

- Group members move to new tables to create a vision and a concrete plan for fostering interdisciplinary collaboration and define next steps, potential partnerships, and future research questions. This activity should respond to the challenges and opportunities identified in the previous section (30 min)
- A whole-group discussion about the entanglement led by the facilitators while developing a summary map based on participants insights - all concept maps are pinned to walls for this activity (20)

**Reflection and next steps (10 min)**

- Facilitators summarise the visions and action plans (5min)

**Participants write down one commitment (5min).**

- This can be e.g., reaching out to another field, exploring a new theory, applying PD in a new way).

## 4 Application to the workshop

To apply for the workshop, please send a one-page abstract to on or before 31 March 2026, briefly describing the case you bring to the workshop and the perspectives and insights that have been pursued when addressing issues of place and spatiality.

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